



Tag Scaling Architecture

1M · 5M · 25M Tags · DigitalOcean Elastic · When to Switch

A company powered by **The Purple Standard**
purple-standard.com

Platform planning · Version 1.0

Document generated: 2026-08-13

Phase 1 baseline · Managed Mongo scalable storage · VM split · sharding

Executive Summary

Contents

- [Executive summary](#)
- [Part 1 — Current architecture \(Phase 1 baseline\)](#)
 - [Topology](#)
 - [Code constraints \(why tags ≠ “just add RAM”\)](#)
 - [Phase 1 numeric limits \(single stack\)](#)
- [Part 2 — Tag definitions & planning formulas](#)
 - [Three “tag” counts](#)
 - [Rollup vs mirror \(operational planning\)](#)
 - [RAM estimate \(TagStore working set\)](#)
 - [Hot Mongo estimate \(7-day @ 5 s sample\)](#)
 - [ScanEngine throughput \(switch trigger\)](#)
- [Part 3 — DigitalOcean elastic services \(how to scale on DO\)](#)
 - [Managed MongoDB — compute + **scalable storage**](#)

mooreVIEW — Tag Scaling Architecture (1M · 5M · 25M)

Current Phase 1 stack · DigitalOcean elastic services · when to switch

Document version: 1.0 · 2026-08-13

A company powered by The Purple Standard · purple-standard.com

Executive summary

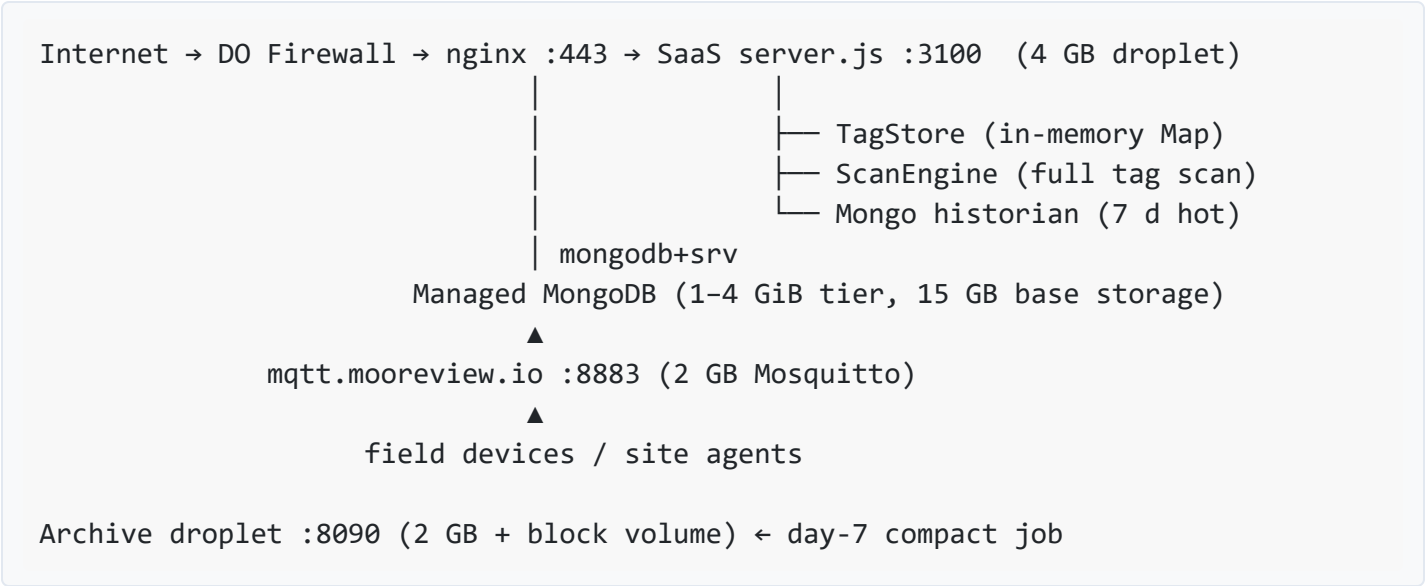
mooreVIEW Cloud today runs as a **Phase 1 monolith** (SaaS + MQTT + archive + Managed Mongo) with a default **MOORE IE MA AG cap of 70,912**. That architecture is correct through **~70K cloud tags** and **~500 cloud-centric field devices**. Beyond that, scale in **stages** — first on **DigitalOcean vertical resize and Managed MongoDB scalable storage**, then **VM split** (`ARCHITECTURE.md`), then **horizontal sharding** — rather than raising one env var on a 4 GB droplet.

Tier	Cloud tags (TagStore)	Architecture mode	Primary DO move
Phase 1	≤ 70,912	Single monolith stack	4 GB SaaS · 15 GB Mongo (baseline)
Phase 2a	70K – 250K	Monolith + tenant shard	8–32 GB SaaS · Mongo 16–32 GB + scalable storage
Phase 2b	250K – 1M	VM split (ingest · GUI · alarms)	Multi-droplet + HA Mongo · Load Balancer
Phase 3	1M – 5M	Sharded ingest + partitioned historian	64 GB Mongo HA · Spaces archive · MQTT cluster
Phase 4	5M – 25M	National platform	Mongo sharding · DOKS · edge rollup mandatory

Critical rule: At **1M+ tags**, **full cloud mirror** of every edge tag is not the default product path — **site-agent rollup** (~15–25% of on-prem tags) keeps cloud TagStore within Phase 2–3 bounds. The **1M / 5M / 25M** rows below model **worst-case full mirror** for capacity planning; operational fleets should plan at **rollup ratios** (see §2).

Part 1 — Current architecture (Phase 1 baseline)

Topology



Code constraints (why tags ≠ “just add RAM”)

Component	File / config	Limiting behavior
TagStore	src/tags/tagStore.js	In-memory Map; one object per tag; assertCapacity() @ MAX_TAGS
Tag cap	MOOREVIEW_MAX_TAGS · src/config.js	Cloud default 70,912 (nextcenturyCloudSizing.js)
ScanEngine	src/runtime/scanEngine.js	Iterates all tags each scan cycle
Historian	src/logger/mongoTagLogger.js	Logs graphEnabled pens; default 5 s sample (MONGODB_SAMPLE_MS)
Hot retention	ARCHIVE_EXPORT.md	7 days Mongo → zstd archive compact
JSON import	REQUEST_JSON_LIMIT=32mb	Large .est tag imports need headroom or chunked merge

Phase 1 numeric limits (single stack)

Resource	Max	Reference fleet
MOORE IE MA AG	70,912	~3K–8K @ 500 devices
Devices / tenant	1,000	500
MQTT connections	~2,000	~532

Historian pens (hot)	~50,000 (Mongo tier bound)	~3,000
Hot Mongo steady	15 GB base plan	~6.5 GB

Detail: `Phase1-SaaS-Architecture-Costs.md` · `runbook`
`docs/CLOUD_DEPLOY_DO_PHASE1_ATL-MQTT.md` .

Part 2 — Tag definitions & planning formulas

Three “tag” counts

Count	Definition	1M example (full mirror)
TagStore tags	Pens in cloud SaaS TagStore	1,000,000
Historian pens	Tags with <code>graphEnabled</code> logged to Mongo	~400,000 @ 40% ratio
MQTT endpoints	Devices/brokers connected to Mosquitto	~200K – 350K @ 3–5 tags/device

Rollup vs mirror (operational planning)

Sync model	Cloud tags @ 667 campuses × 1,500 assets	Use
Full mirror (~3 tags/asset)	~3,000,000	Avoid — forces Phase 4
Rollup (~800 pens/campus)	~534,000	Recommended — Phase 2b
Per-campus tenant (~4,500/campus)	4,500 × N tenants	Franchise / multi-org

RAM estimate (TagStore working set)

Directional: ~1.5 – 2.5 KB / tag in Node heap (metadata + runtime state + average function-block footprint).

Tags	Est. TagStore RAM	Min SaaS RAM (TagStore only)
70,912	~140 MB	4 GB (Phase 1 OK)
250,000	~500 MB	8 GB
1,000,000	~2 GB	16 GB+ (scan overhead dominates)
5,000,000	~10 GB	Split ingest — no single process
25,000,000	~50 GB	Sharded — not one TagStore

Hot Mongo estimate (7-day @ 5 s sample)

Calibrated from Phase 1 reference: ~3,000 pens → ~6.5 GB hot ⇒ ~2.2 MB / pen / 7 days.

TagStore tags	Historian pens (40%)	Hot Mongo (7 d)	DO Mongo action
70,912	~28,000	~ 60 GB	16 GB tier + scalable storage
250,000	~100,000	~ 220 GB	32 GB tier + storage
1,000,000	~400,000	~ 880 GB	64 GB HA + ~900 GB scalable storage
5,000,000	~2,000,000	~ 4.4 TB	64 GB+ HA + multi-TiB storage or historian partition
25,000,000	~10,000,000	~ 22 TB	Sharded cluster · tiered hot/cold · Spaces archive

Reduce pens via longer sample interval, rollup-only sync, or edge historian for ALF/apartment/hotel fleets.

ScanEngine throughput (switch trigger)

Monolith scan cost grows **O(n tags)**. Directional sustained limits on a single Node process:

SaaS droplet	Practical tag scan ceiling @ 1 s scan	Action if exceeded
4 GB	~ 50K – 80K	Step RAM; reduce scan rate
16 GB	~ 150K – 250K	VM split — dedicated ingest
32 GB	~ 250K – 400K (soft)	Partition by tenant/site
Multi-node ingest	1M+	Sharded ScanEngine (platform work)

Part 3 — DigitalOcean elastic services (how to scale on DO)

Managed MongoDB — compute + scalable storage

Primary elastic database service for hot historian and tenant config.

Feature	Use for mooreVIEW	Notes
Vertical resize	Step 1 → 4 → 16 → 32 → 64 GiB RAM tiers	Brief failover on HA clusters
Scalable storage	Add 10 GiB increments @ ~ \$0.215/GiB/mo	Scale storage without compute — ideal for historian growth
Storage autoscale	Enable @ ~ 80% disk threshold	Adds increment per node on HA clusters
3-node HA replica set	Production @ ≥ 1M tags	~3× single-node list price
Max storage	Up to ~ 16 TiB per cluster (plan-dependent)	See DO Mongo plan storage range

List pricing (directional, 2026): **1 GiB \$15/mo · 4 GiB \$61/mo · 16 GiB \$244/mo · 32 GiB \$488/mo · 64 GiB \$975/mo.**

Switch to scalable storage when hot Mongo exceeds **base plan included GiB** (~15 GB on entry tier) — **before** jumping compute tier if CPU is idle.

Droplets — vertical scale (SaaS · MQTT · archive)

Workload	Phase 1	Phase 2a	Phase 2b / 1M	Phase 3 / 5M
SaaS / GUI	4 GB (\$24)	8–16 GB (\$48–96)	2–4 × 16–32 GB + LB	DOKS pool or 8+ dedicated nodes
MQTT broker	2 GB (\$12)	8 GB (\$48)	2 × 8 GB cluster	EMQX/HiveMQ or 3-node Mosquitto
Archive API	2 GB + volume	4 GB + volume	Spaces primary	Spaces Cold for compliance years

Memory-optimized droplets (directional): **64 GiB / 8 vCPU ~\$524/mo · 128 GiB / 16 vCPU ~\$1,048/mo** — use for **single-tenant ingest** only until DOKS split.

Spaces object storage — archive tier

Replace block-volume archive when cumulative zstd exceeds **~500 GB – 1 TB**.

Item	Rate
Base	\$5/mo (250 GiB + 1 TiB egress)
Additional Standard	\$0.02/GiB/mo
Cold storage	\$0.007/GiB/mo (13 mo compliance)

Switch from archive droplet → Spaces when block volume > **500 GB** or multi-TB fleet (**Phase1-SaaS-Architecture-Costs.md** trigger).

Load Balancers & DOKS

Service	When to add	Cost (directional)
Load Balancer	≥ 2 SaaS GUI nodes (Phase 2b)	~\$12/mo
DOKS (Kubernetes)	≥ 5M tags or > 10 SaaS replicas	Control plane + wor c

- 5. **Migrate archive to Spaces** before **500 GB** block volume
- 6. **Step Mongo compute tier** when query latency or CPU bound (not just disk)
- 7. **Enable Mongo HA 3-node** before **1M tags** production
- 8. **Introduce DOKS / sharding** at **5M+ tags** — do not vertical-scale past **128 GB** monolith

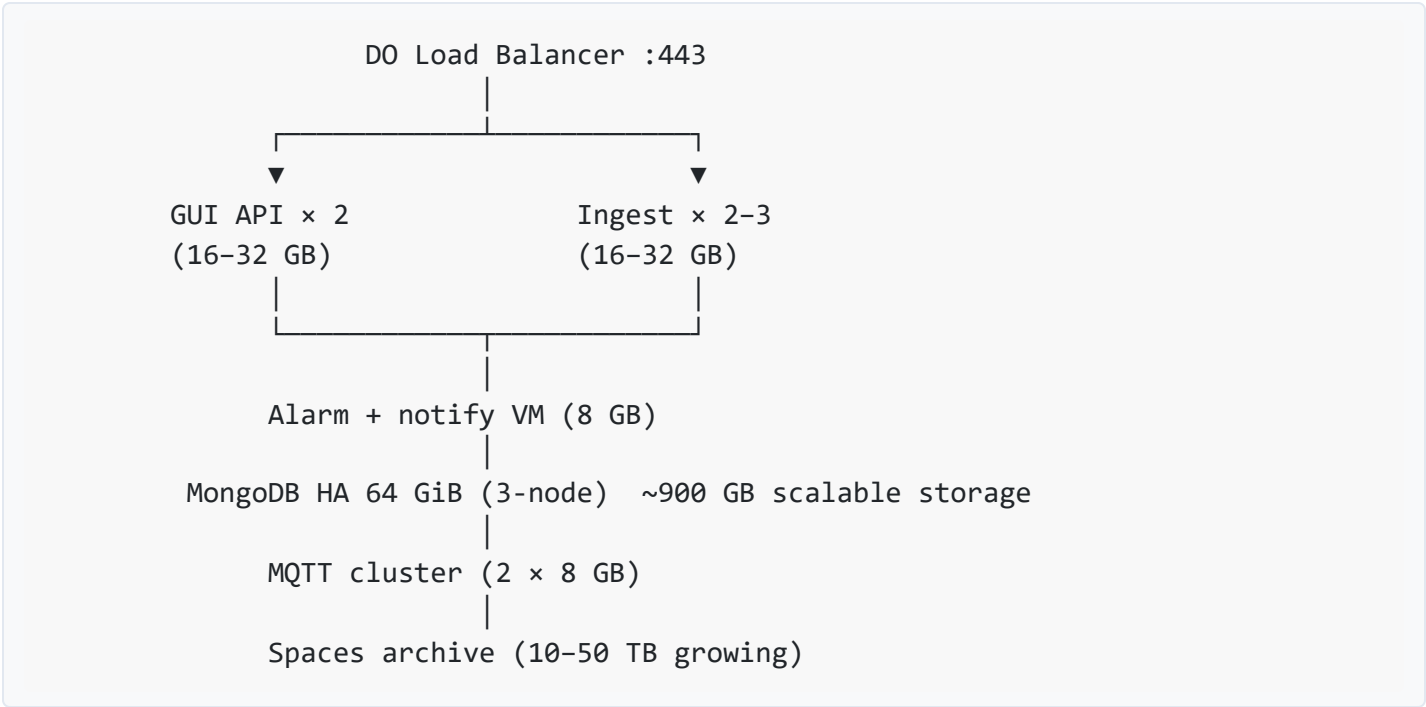
Part 4 — Tier detail @ 1M · 5M · 25M tags

Assumptions (full-cloud mirror worst case)

Parameter	Value
Historian pen ratio	40% of TagStore tags
Sample interval	5 s
Hot retention	7 days
Archive	zstd compact → Spaces @ Phase 3+
Region	NYC1 / NYC3

@ 1,000,000 tags — Phase 2b → early Phase 3

Architecture



Resource	Specification
----------	---------------

TagStore	2–4 shards by tenantId or siteId (platform) — not one 1M Map
MOORE IE MA AG	262,144 – 1,048,576 per shard
Hot Mongo	~880 GB — 64 GiB HA + ~900 GiB scalable storage
MQTT	~200K – 350K connections → 2-node broker cluster
Scan	Partitioned ingest — ≤ 250K tags / process

When to switch TO this tier

Signal	Threshold
TagStore count	> 250,000 on one tenant or aggregate
SaaS RAM	> 70% sustained on 32 GB droplet
Scan cycle	> 2 s p95 @ 1 s target
Hot Mongo	> 200 GB steady
MQTT	> 2,000 connections on 2 GB broker

Directional DO opex (cloud only)

Line	Monthly
SaaS 4 × 16 GB ingest/GUI/alarm	~\$384
MQTT 2 × 8 GB	~\$96
Mongo 64 GiB HA × 3	~\$2,925
Mongo ~900 GiB extra storage	~\$194
Spaces ~20 TB archive	~\$410
Load balancer	~\$12
Total	~\$4,000 – 5,000/mo

Platform work required: VM split per ARCHITECTURE.md · tag partition · message bus for alarm:transition · historian index by {tagId, ts} · archive writer to Spaces.

@ 5,000,000 tags — Phase 3

Architecture

DOKS or dedicated fleet

- └ ingest deployment (N replicas, HPA on MQTT lag)
- └ gui-api deployment (M replicas)
- └ alarm-notify deployment
- └ archive-compactor CronJob → Spaces

MongoDB: sharded cluster (shard key: tenantId + siteId)

- └ shard A: 0–1.25M tags
- └ shard B: 1.25–2.5M
- └ shard C: 2.5–3.75M
- └ shard D: 3.75–5M

Hot tier: 7 d rolling per shard · ~4.4 TB aggregate

Cold tier: Spaces Standard → Cold @ 13 mo

MQTT: dedicated cluster (3+ nodes) or managed EMQX

Edge policy: rollup required – mirror only by exception

Resource	Specification
TagStore	Mongo-backed or Redis with local LRU — not full in-memory Map
Hot Mongo	~4.4 TB — multiple 64 GiB shards or storage-optimized DO plans
Historian	Partition collections by week · consider downsample after 24 h
Ingest	Horizontally scaled · 10M device path in ARCHITECTURE.md

When to switch TO this tier

Signal	Threshold
Aggregate tags	> 1,000,000 full mirror OR > 5,000,000 rollup-equivalent load
Mongo	Single cluster > 1 TiB hot or CPU > 60% sustained
Ingest	> 50K MQTT msg/s peak
SaaS nodes	> 8 vertical-scaled droplets — move to DOKS

Directional DO opex

Line	Monthly
Compute (DOKS 8 × 16 GB workers + 3 × 32 GB dedicated)	~\$2,500 – 4,000
Mongo sharded (4 × 64 GiB HA)	~\$12,000 – 15,000
Mongo / Spaces storage ~ 5 – 8 TB hot + 100+ TB archive	~\$2,000 – 4,000
MQTT cluster / LB / monitoring	~\$500 – 800

Total	~\$17,000 – 25,000/mo
-------	-----------------------

Platform work required: Shard router · scan worker pool · stream bus (Kafka/Redis Streams) · tenant quota enforcement · **mandatory rollup** for campus verticals.

@ 25,000,000 tags — Phase 4 (national ceiling)

Architecture

Aligns with MooreVIEW-Infrastructure-Projections.md Part 7 ceiling and ARCHITECTURE.md 10M devices / 10K users target — extended to 25M tag pens (multi-vertical national fleet).

Multi-region DO (NYC + ATL + SFO) – tenant-aware routing

- Regional ingest pools (DOKS)
- Global GUI + auth (CDN-fronted)
- Alarm/notify regional fanout
- Archive: Spaces multi-bucket · PB-class · lifecycle to Cold

MongoDB: 8–16 shards · 64–128 GiB primaries · TiB+ scalable storage each

Historian: hot 7 d · warm 90 d Mongo · cold Spaces · compliance 13 mo edge

Edge: 100K+ IoT-Link + 500+ MVP Suite – **rollup only to cloud**

Resource	Specification
TagStore	Distributed — no single-process TagStore
Hot Mongo	~22 TB theoretical @ full mirror — reduce to ~2–4 TB via rollup + downsample
Archive	PB-class Spaces · \$0.007/GiB Cold for year 2+
MQTT	Regional brokers · 100K+ connections per region

When to switch TO this tier

Signal	Threshold
National fleet	> 100K sites or > 5M tags sustained with full mirror
Single-region DO	Egress / latency SLO breach
Mongo	> 5M tags hot historian without shard expansion
Business	\$50M+ ARR portfolio (Infrastructure Projections ceiling)

Directional DO opex

Line	Monthly
------	---------

Multi-region compute (DOKS + dedicated)	~\$15,000 – 25,000
Mongo sharded national	~\$50,000 – 80,000
Storage (hot + warm + 500 TB – 2 PB archive)	~\$10,000 – 40,000
Network / LB / observability	~\$3,000 – 5,000
Total	~\$80,000 – 150,000/mo

Contact DO sales for storage-optimized and multi-TiB Mongo commitments.

Part 5 — When to switch (decision matrix)

By tag count (primary trigger)

Cloud TagStore tags	Stay on	Switch to	DO elastic action
≤ 70,912	Phase 1 monolith	—	4 GB SaaS · 15 GB Mongo
70K – 131K	Phase 1 (stretched)	Phase 2a	8–16 GB SaaS · MAX_TAGS=131072 · Mongo 16 GB + storage
131K – 250K	Phase 2a	Phase 2a shard	2 × 16 GB SaaS OR per-tenant VM · Mongo 32 GB
250K – 1M	—	Phase 2b VM split	LB + Mongo HA 32–64 GB · scalable storage · Spaces archive
1M – 5M	Phase 2b	Phase 3 sharded	DOKS · Mongo sharding · MQTT cluster
5M – 25M	Phase 3	Phase 4 national	Multi-region · PB Spaces · edge rollup policy

By resource signal (override tag count)

Signal	Threshold	Switch
MOOREVIEW_MAX_TAGS API 413 errors	Any	Raise cap or shard tenant before raising cap
SaaS heap > 80%	Sustained 15 min	Vertical resize then split ingest
Mongo disk > 80%	Primary node	Enable autoscale storage (10 GiB increment)
Mongo CPU > 60%	Sustained	Resize compute tier (not just storage)
MQTT connections	> 2,000	Second broker · bridge · or EMQX cluster
Archive volume	> 500 GB	Spaces migration
p95 API latency	> 500 ms on tag list	Partition TagStore · CDN static · WS throttling

Full mirror requested @ > 18 campuses	~81K+ tags	Reject — enforce model B rollup
--	------------	---

Vertical-specific shortcuts

Vertical	Tags @ scale	Cloud plan
Wastewater 500 devices	~7K	Phase 1 — no change
ALF 60 campuses	~48K–72K rollup · 270K on-prem	Phase 2 SaaS · rollup only
Apartment 20 / Hotel 12	~16K–24K / ~10K–14K	Phase 1
Enterprise full mirror	1M+	Phase 2b minimum

Part 6 — Migration runbook (Phase 1 → 2b @ ~250K tags)

Ordered steps on DigitalOcean **without downtime goal**:

1. **Provision Mongo HA 32 GiB** cluster · enable **scalable storage** + **autoscale @ 80%**
2. **Migrate hot historian** (`mongodump` / logical migration) · verify indexes on `{ tagId: 1, ts: -1 }`
3. **Deploy second SaaS ingest node** — MQTT consumer only · same codebase
`MOOREVIEW_ROLE=ingest`
4. **Add DO Load Balancer** — GUI node registers; ingest bypasses LB (MQTT direct)
5. **Split alarm notify** to third VM · wire `eventBus` → Redis/Kafka (platform seam)
6. **Raise MOORE IE MA AG** per shard · **never** above **250K** per process
7. **Migrate archive** to **Spaces** · point `ARCHIVE_SERVER_URL` to S3-compatible adapter
8. **Load test @ 80%** target tags · watch scan p95 · Mongo disk slope
9. **Cutover DNS** · drain Phase 1 monolith
10. **Document tenant shard map** in `cloud_tenants.json` / Mongo `tenants` collection

Rollback: keep Phase 1 stack **7 days** · DNS revert · Mongo read-only replica of old cluster.

Part 7 — Cost vs revenue sanity (directional)

Cloud opex alone — **not** cellular or support labor.

Tags	Est. cloud opex/mo	Illustrative renewal to cover cloud @ 10% rule
70,912	~\$150 – 300	~\$1.5K – 3K/mo
250,000	~\$800 – 1,500	~\$8K – 15K/mo
1,000,000	~\$4K – 5K	~\$40K – 50K/mo

5,000,000	~\$17K – 25K	~\$170K – 250K/mo
25,000,000	~\$80K – 150K	~\$800K – 1.5M/mo

At ALF/apartment pricing (\$110/asset), 1M cloud tags (rollup) is **far smaller** than 1M mirror — operational fleets should size cloud tags, not on-prem asset counts.

Appendix — related documents

Document	Purpose
Phase1-SaaS-Architecture-Costs.md	Phase 1 limits & 500-device opex
D0-Phase1-Storage-Cost-Review.md	Mongo storage economics
docs/ARCHITECTURE.md	VM split · 10M device target
docs/ARCHIVE_EXPORT.md	Hot → archive pipeline
MooreVIEW-Infrastructure-Projections.md	Y1–Y5 + Part 10–11 verticals
DO Mongo scalable storage	Autoscale & resize
DO Spaces pricing	Archive tier

Env vars (scaling-relevant)

Variable	Default	Scale note
MOOREVIEW_MAX_TAGS	70912 (cloud)	Per-process cap — shard above 250K
MOOREVIEW_DEPLOYMENT	c <code>loud</code>	Enables cloud sizing defaults
MONGODB_URI	—	Point to HA cluster @ 1M+
MONGODB_SAMPLE_MS	5000	Increase to reduce hot Mongo slope
ARCHIVE_SERVER_URL	archive droplet	Spaces @ TB scale

Disclaimer: Directional capacity planning — not an SLA, quote, or commitment to implement sharding by a specific date. Code changes for Phase 2b+ are roadmap items in `ARCHITECTURE.md` .

Tag Scaling Architecture v1.0 — mooreVIEW platform planning.

mooreVIEW is a company powered by [The Purple Standard](#). © Purple Standard Holdings.